PS-113

SCHOOL FEE MANAGEMENT ENGINE

WEEK 1 – SOFTWARE DESIGN DOCUMENT

Domain	Public Services – School Administration
Technology	Java 17+
Project Type	Command-file-driven application (No UI)
Submission	Week 1 – Understand and Design
Team Members	_____
College	_____
Academic Year	2026–2027

1. Introduction

The School Fee Management Engine is a software system designed to automate school fee calculations and payment management. The system calculates annual fees based on grade-wise fee structures, applies eligible concessions, generates instalment schedules, manages transport and hostel charges, calculates late fees and maintains payment records.

It also handles mid-year admissions, student withdrawals, refunds and fee clearance. The application uses input files such as CSV files to process student details, fee structures, add-ons and payments. It generates fee demands, student ledgers and defaulter reports.

2. Objectives

• Automate annual school fee calculations.

• Apply sibling and category-based concessions.

• Generate annual, term and monthly instalment schedules.

• Manage transport and hostel charges.

• Calculate late fees and outstanding dues.

• Process payments and bounced cheques.

• Handle mid-year admissions and withdrawals.

• Generate fee demand, ledger and defaulter reports.

3. Scope of the Project

• Student registration and sibling management.

• Grade-wise fee structure management.

• Concession calculation and application.

• Instalment generation.

• Transport and hostel fee management.

• Payment processing and reconciliation.

• Late fee calculation.

• Admission and withdrawal processing.

• Refund and Transfer Certificate (TC) clearance management.

• Report generation.

The system will not contain a graphical user interface. All operations will be performed using command files.

4. Glossary of Terms

Term	Description
Fee Head	Individual fee category such as tuition, lab and exam.
Concession	Reduction in the payable fee.
Sibling Concession	Discount based on the number of siblings enrolled.
Category Concession	Discount based on staff ward, merit or EWS category.
Instalment	A portion of the total fee paid on a particular date.
Pro-rating	Calculating fees according to the remaining academic months.
Add-on	Additional transport or hostel charges.
Late Fee	Additional charge for delayed payment.
Ledger	Record of fees, payments and outstanding balances.

Term	Description
Advance Credit	Extra payment available for future instalments.
Defaulter	Student with dues outstanding beyond 60 days.
Security Deposit	Refundable amount collected from hostel students.
Fee Demand	Amount payable by a student.
TC	Transfer Certificate issued after fee clearance.
Fee Revision	Modification of the existing fee structure.

5. System Architecture

The application is divided into modules. Each module performs a specific operation.

Module	Responsibility
Student Management	Stores student, grade, section and sibling details.
Fee Structure	Stores grade-wise fee details.
Concession Management	Calculates category and sibling concessions.
Instalment Management	Generates payment schedules.
Add-on Management	Calculates transport and hostel charges.
Payment Management	Records payments and cheque bounces.
Late Fee Management	Calculates late charges and dues.
Withdrawal Management	Calculates refunds and checks TC eligibility.
Report Management	Generates demands, ledgers and defaulter reports.

Processing Flow

Input CSV Files → Student Management → Fee Structure → Concession Calculation → Instalment Generation → Transport and Hostel Charges → Payment Processing → Late Fee Calculation → Ledger Management → Report Generation → Fee Demand / Ledger / Defaulter List

6. Class Diagram

The system follows Object-Oriented Programming (OOP). Each class represents an important entity or operation.

Class	Responsibility
Student	Maintains student information.
FeeStructure	Maintains grade-wise fee details.
FeeHead	Stores individual fee categories.
Concession	Calculates eligible discounts.
Instalment	Maintains instalment amounts and due dates.
Payment	Stores payment information.
AddOn	Manages additional charges.
Transport	Calculates route-based transport charges.
Hostel	Calculates hostel fees and deposits.
Ledger	Maintains financial transactions.
FeeEngine	Performs the main fee calculations.
Withdrawal	Manages refunds and TC clearance.
ReportGenerator	Generates required reports.

UML Class Structure (text representation)

Student: studentId, name, grade, section, admissionDate, status
FeeStructure: grade; getFeeDetails(); reviseFee()
FeeHead: headName, amount
Concession: category, percentage; calculateDiscount()
Instalment: instalmentId, dueDate, amount, paidAmount, status
Payment: paymentId, paymentDate, amount, paymentMode, status
AddOn: type, amount, startDate, endDate
Transport extends AddOn: route; calculateTransportFee()
Hostel extends AddOn: roomType, securityDeposit; calculateHostelFee()
Ledger: totalFee, totalPaid, balance; generateLedger()
FeeEngine: calculateAnnualFee(), applyConcessions(), generateInstalments(), applyPayment(), calculateLateFee()
Withdrawal: noticeDate; calculateRefund(), checkTCClearance()
ReportGenerator: generateFeeDemand(), generateLedger(), generateDefaulterList()

Relationships: Student is associated with FeeStructure, Concession, Instalment, Payment, AddOn, Ledger and Withdrawal. FeeStructure contains FeeHead. Transport and Hostel extend AddOn. FeeEngine processes Student, Instalment and Ledger. ReportGenerator uses Ledger.

7. State Diagram

The state diagram explains the different stages of a student's fee lifecycle.

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ENROLLED
↓
FEE GENERATED
↓
UNPAID
■■■→ PARTIALLY PAID ■■→ FULLY PAID
■■■→ DEFAULTER ■■→ PARTIALLY PAID / FULLY PAID

ENROLLED / UNPAID / PARTIALLY PAID / FULLY PAID
↓
WITHDRAWAL REQUESTED
■■■→ TC BLOCKED (outstanding dues)
■■■→ TC ELIGIBLE (no dues)
TC BLOCKED ■■ dues cleared ■■→ TC ELIGIBLE
TC ELIGIBLE ■■ TC issued ■■→ WITHDRAWN
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Explanation of States

State	Description
Enrolled	Student is registered.
Fee Generated	Student's annual fee is calculated.
Unpaid	Fee is generated but not paid.
Partially Paid	Some amount is paid.
Fully Paid	All applicable dues are cleared.
Defaulter	Dues have crossed 60 days.
Withdrawal Requested	Student requests withdrawal.
TC Blocked	Outstanding dues remain.
TC Eligible	All dues are cleared.
Withdrawn	Student has left the school.

8. Business Rules

The following business rules define how the fee management engine must operate.

8.1 Fee Structure and Instalment Rules

Rule ID	Business Rule
BR-01	Each grade must have a separate fee structure.
BR-02	Fee structure must contain tuition, development, lab, activity and exam fees.
BR-03	Annual plan must provide a 3% discount on tuition.
BR-04	Annual plan must generate one instalment.

Rule ID	Business Rule
BR-05	Term plan must divide the fee into 40%, 30% and 30%.
BR-06	Term instalment due dates must be 10 April, 10 August and 10 December.
BR-07	Monthly plan must generate 10 instalments from April to January.
BR-08	Monthly instalments must be equally divided, with the January instalment absorbing rounding differences.
BR-09	Instalment plan can be changed only before the first due date.

8.2 Concession Rules

Rule ID	Business Rule
BR-10	Second enrolled child receives 10% sibling concession on tuition.
BR-11	Third and subsequent enrolled children receive 20% sibling concession on tuition.
BR-12	Eldest enrolled child receives no sibling concession.
BR-13	Staff ward receives 50% tuition concession.
BR-14	Merit students scoring at least 90% in the previous year receive 25% tuition concession.
BR-15	EWS students receive 100% tuition and 50% concession on other eligible heads.
BR-16	Only the highest applicable category concession can be selected.
BR-17	Sibling concession must be added to one selected category concession.
BR-18	Combined tuition concession must not exceed 60%.
BR-19	If the eldest sibling withdraws, sibling rankings must be recalculated from the next instalment.
BR-20	Plan discount must be applied after concession calculation.

8.3 Transport and Hostel Rules

Rule ID	Business Rule
BR-21	Transport charges must depend on the selected route.
BR-22	Zone A transport fee is ₹1,800 per month.
BR-23	Zone B transport fee is ₹2,400 per month.
BR-24	Zone C transport fee is ₹3,000 per month.
BR-25	Transport joined on or before the 15th must be charged for the full month.
BR-26	Transport joined after the 15th must be charged at 50% for that month.
BR-27	Transport charges must be billed according to the months covered by the instalment.
BR-28	Shared hostel fee must be ₹9,000 per month.
BR-29	Single hostel fee must be ₹12,000 per month.
BR-30	Hostel security deposit must equal one month's hostel fee and be collected once.
BR-31	Security deposit must be refunded after deducting applicable damages.
BR-32	Hostel fees must not receive any concession.

8.4 Late Fee and Payment Rules

Rule ID	Business Rule
BR-33	Late fee must be ₹50 per day after the applicable due date.
BR-34	Maximum late fee must be ₹1,500 per instalment.
BR-35	April instalment must have a grace period of 7 days.
BR-36	No April late fee must be charged during the grace period.
BR-37	Students with dues outstanding beyond 60 days must be classified as defaulters.
BR-38	Defaulters must receive a WITHHELD exam hall-ticket flag.

Rule ID	Business Rule
BR-39	Payments must first be applied to late fees.
BR-40	Payments must next cover the oldest instalment's tuition.
BR-41	Remaining payment must cover other heads of that instalment.
BR-42	Add-on charges must be paid after the other heads.
BR-43	Remaining payment must be applied to the next instalment.
BR-44	Overpayment must be stored as advance credit.
BR-45	Bounced cheques must incur a ₹500 charge.
BR-46	Bounced payments must be reversed using the original payment date.
BR-47	Late fees must be recalculated after cheque reversal.

8.5 Admission and Withdrawal Rules

Rule ID	Business Rule
BR-48	Mid-year tuition and eligible other heads must be pro-rated based on remaining months.
BR-49	Admission on or before the 15th must count the current month.
BR-50	Admission after the 15th must exclude the current month.
BR-51	Admission fee must be a flat ₹10,000.
BR-52	Development fee must never be pro-rated.
BR-53	Tuition refund must be allowed only for full unstarted terms.
BR-54	Withdrawal notice must be provided at least 30 days before the term start.
BR-55	Other fee heads must be non-refundable.
BR-56	Withdrawal during the last term must not receive any refund.
BR-57	TC must be issued only when all outstanding dues are zero.

8.6 Fee Revision and Reporting Rules

Rule ID	Business Rule
BR-58	Mid-year fee revision up to 10% must apply only to instalments not yet due.
BR-59	Fee revision above 10% must require BOARD_APPROVED status.
BR-60	Fee revision above 10% must apply only from the next academic year.
BR-61	System must generate fee demand reports.
BR-62	System must generate student-wise ledgers.
BR-63	System must generate defaulter lists for a selected date.
BR-64	Every payment, refund and adjustment must be recorded in the ledger.

9. Important Edge Cases

No.	Edge Case	Expected Result
1	Staff ward is also the third sibling	50% category + 20% sibling = 70%, capped at 60%.
2	Transport joined on 15th	Full monthly transport fee.
3	April payment on 7th day of grace	No late fee.
4	Eldest sibling withdraws mid-term	Re-ranking starts from the next instalment.
5	Bounced cheque after late fee reaches ₹1,500	Reverse payment and recompute late fee, retaining the cap.
6	Mid-year admission on 16th	Current month excluded from pro-rating.
7	Notice exactly 30 days before term	Eligible for refund if the term is unstarted.

No.	Edge Case	Expected Result
8	Monthly plan rounding difference	January instalment absorbs the difference.

10. Assumptions and Design Considerations

• All monetary values will be calculated in Indian Rupees (₹).

• Dates will follow the DD-MM-YYYY format in reports.

• Each student will have a unique student ID.

• All fee transactions will be maintained in the student's ledger.

• Concession calculations will follow the specified precedence.

• The combined category and sibling tuition concession will be capped at 60%, before applying the annual plan discount.

• Hostel security deposits will be maintained separately from regular hostel charges.

• A payment reversal will restore the outstanding amount and trigger late-fee recalculation.

• The seven-day April grace period will be treated as seven calendar days following the due date.

• Defaulter status will be evaluated using the outstanding instalment's due date and applicable unpaid balance.

11. Expected Output

Output	Description
Fee Demand	Student-wise total payable amount.
Instalment Schedule	Instalment amounts and due dates.
Student Ledger	Complete fee and payment history.
Defaulter List	Students with dues beyond 60 days.
Payment Report	Successful, reversed and bounced payments.
Withdrawal Report	Refund calculation and TC clearance status.

12. Conclusion

The School Fee Management Engine aims to simplify and automate the fee management process in schools. It reduces manual calculations, improves accuracy in concession handling, maintains transparent payment records and helps school administrators manage student fees efficiently. The proposed design uses Object-Oriented Programming concepts to make the application modular, reusable and easy to maintain.

13. Week 1 Submission Checklist

☐ Introduction and objectives

☐ Scope of the project

☐ Glossary of terms

☐ System architecture

☐ UML class diagram

☐ Student fee lifecycle state diagram

☐ Business rules BR-01 to BR-64

☐ Edge cases

☐ Assumptions and expected outputs

☐ Conclusion